

Ectoprocta (4,500 species)



Ectoprocts

Ectoprocts (also known as bryozoans) live as sessile colonies and are covered by a tough exoskeleton (see Concept 33.3).

Gastrotricha (800 species)

Gastrotrichs are tiny worms whose ventral surface is covered with cilia, leading them to be called hairy bellies. Most species live on the bottoms of lakes or oceans, where they feed on small organisms and partially decayed organic matter. This individual has consumed algae, visible as the greenish material inside its gut.



A gastrotrich (differential interference contrast LM)

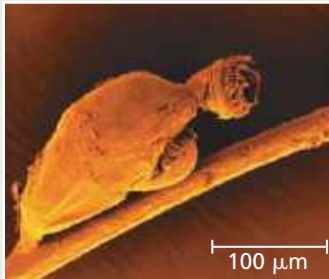
Brachiopoda (335 species)



A brachiopod

Brachiopods, or lamp shells, may be easily mistaken for clams or other molluscs. However, most brachiopods have a unique stalk that anchors them to their substrate, as well as a crown of cilia called a lophophore (see Concept 33.3).

Cycliophora (1 species)



A cycliophoran (colorized SEM)

The only known cycliophoran species, *Symbion pandora*, was discovered in 1995 on the mouthparts of a lobster. This tiny creature has a unique body plan and a bizarre life cycle: Males impregnate females that are still developing in their mothers' bodies. The fertilized females escape, settle elsewhere on the lobster, and release their offspring.



A ribbon worm

Nemertea (900 species)

Also called ribbon worms, nemerteans swim through water or burrow in sand, extending a unique proboscis to capture prey. Nemerteans have a reduced coelom, so their bodies are relatively solid. They also have an alimentary canal and a closed circulatory system in which the blood is contained in vessels and hence is distinct from fluid in the body cavity.

Annelida (16,500 species)

Annelids, or segmented worms, are distinguished by their body segmentation. Earthworms are the most familiar annelids, but most annelids live in marine or freshwater habitats (see Concept 33.3).



A marine annelid

Mollusca (100,000 species)

Molluscs (including snails, clams, squids, and octopuses) have a soft body that in many species is protected by a hard shell (see Concept 33.3).



An octopus

Ecdysozoa

Loricifera (10 species)

Loriciferans (from the Latin *lorica*, corset, and *ferre*, to bear) are tiny animals that inhabit sediments on the seafloor. A loriciferan can telescope its head, neck, and thorax in and out of the lorica, a pocket formed by six plates surrounding the abdomen. Though the natural history of loriciferans is mostly a mystery, at least some species likely eat bacteria.



A loriciferan (LM)

Priapulida (16 species)



A priapulid

Priapulans are worms with a large, rounded proboscis at the anterior end. (They are named after Priapos, the Greek god of fertility, who was symbolized by a giant penis.) Ranging from 0.5 mm to 20 cm in length, most species burrow through seafloor sediments. Fossil evidence suggests that priapulans were among the major predators during the Cambrian period.